

Bellhouse Endowment

The optimizer came back with a 341 percent long position and a 190 percent short, and the analyst who ran it sent the output to the investment committee without comment.

Bellhouse College's endowment was \$1.9 billion. The investment office had hired a new director of research, Nnamdi Achebe, who had spent his first month rebuilding the asset allocation work in code rather than in the consultant's spreadsheet. He had asked the AI environment for a general mean-variance optimizer — N assets, a covariance matrix, minimum-variance and tangency portfolios, with and without short-sale constraints — and had it working in under an hour.

He fed it twenty asset classes and sixty months of returns. The unconstrained tangency portfolio it produced held 341 percent of the endowment in emerging market small-cap value, financed by a 190 percent short in US large-cap growth, with fourteen of the twenty asset classes carrying weights between negative 40 and positive 60 percent. The portfolio's in-sample Sharpe ratio was 1.83, which was roughly twice what any diversified institution had achieved over any long period.

Achebe did not believe the output. He also could not immediately say what was wrong with it, because the code was correct — he had checked the matrix algebra line by line, and it did what mean-variance optimization is supposed to do.

He decided to test it on a case small enough to check by hand. Two assets, where the minimum-variance weight has a closed form that fits on one line and a student can compute it on paper. He picked the two largest allocations in Bellhouse's actual portfolio: domestic equity, with a standard deviation of 18 percent, and developed international equity, with a standard deviation of 22 percent. The correlation between them over the sixty months was 0.85.

The closed form gave a weight of 109.4 percent in domestic equity and negative 9.4 percent in international. The code agreed to six decimals.

Achebe then changed the correlation to 0.90 — inside any reasonable confidence interval on a sixty-month estimate — and the weight moved to 134.0 percent.

EXHIBIT 1 — TWO-ASSET TEST CASE

	DOMESTIC EQUITY	DEVELOPED INTERNATIONAL
Expected excess return	5.5%	4.0%
Standard deviation	18.0%	22.0%

Correlation estimated over 60 monthly observations: 0.85. The standard error on a correlation estimated from 60 observations at $\rho = 0.85$ is approximately 0.035.

Bellhouse's policy portfolio, set by the committee in 2021, holds 32 percent domestic equity, 18 percent developed international, 12 percent emerging markets, 20 percent private equity, 10 percent real assets, and 8 percent fixed income. It has never held a short position.

ASSIGNMENT

- 1 Compute the minimum-variance portfolio of the two assets by hand, using the closed form. Confirm it against code you write or generate. Report both.
- 2 Recompute at correlations of 0.75, 0.85, and 0.90. Report the weights and the resulting portfolio standard deviations.
- 3 The portfolio standard deviation barely changes across those three correlations while the weights change by 46 percentage points. What does that tell you about the shape of the objective function near its minimum, and what does it imply for how much confidence to place in an optimizer's weights?
- 4 Compute the tangency weights at $\rho = 0.85$, then recompute holding everything fixed except the expected excess return on domestic equity, at 5.0, 5.5, and 6.0 percent. Report the weights.
- 5 Achebe's twenty-asset output is not a coding error. Say what it is, and recommend what Bellhouse should do — specifically enough that the recommendation could be implemented on Monday.